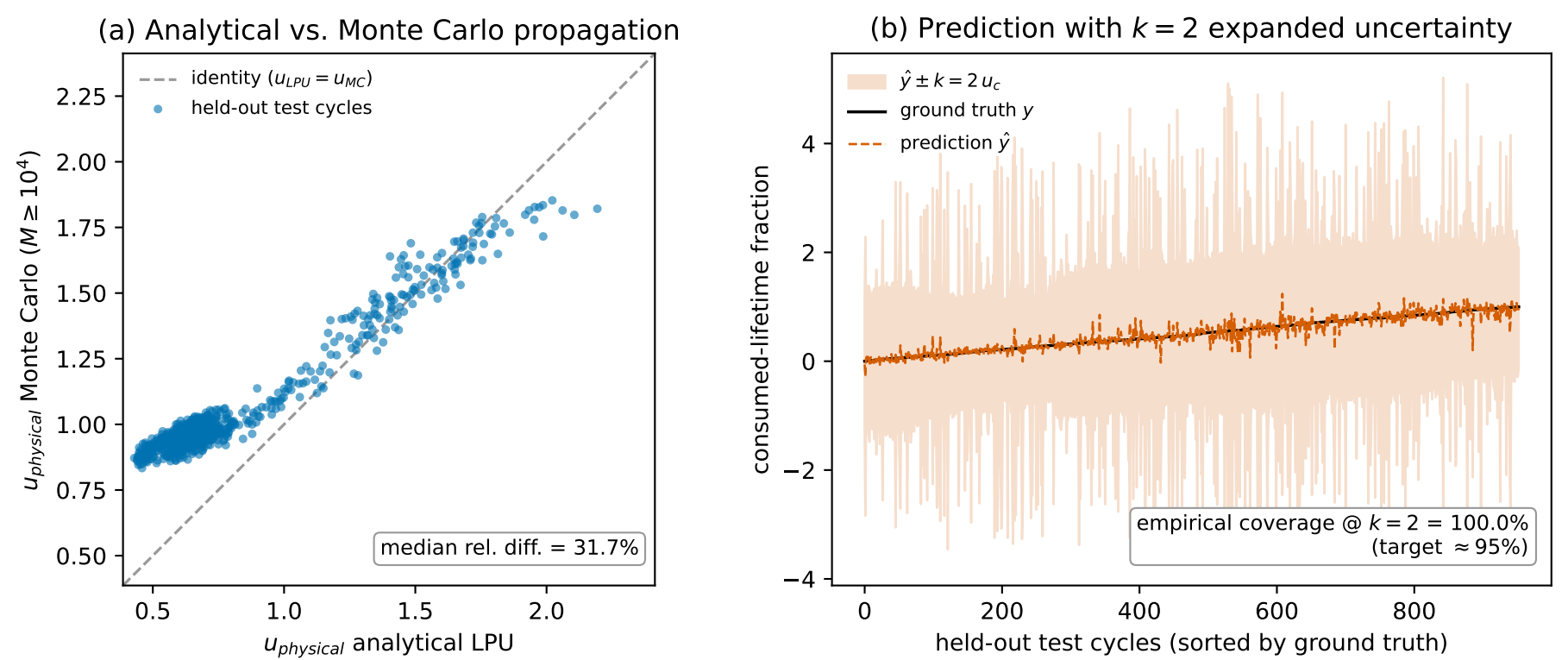

(a) Analytical vs. Monte Carlo propagation
identity ($u_{LPU} = u_{MC}$)
held-out test cycles
median rel. diff. = 31.7%
$u_{physical}$ Monte Carlo ($M \geq 10^4$)
$u_{physical}$ analytical LPU
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2.25
0.5
1.0
1.5
2.0
(b) Prediction with $k = 2$ expanded uncertainty
$\hat{y} \pm k = 2\,u_c$
ground truth y
prediction $\hat{y}$
empirical coverage @ $k = 2$ = 100.0%
(target ≈ 95%)
consumed-lifetime fraction
held-out test cycles (sorted by ground truth)
−4
−2
0
2
4
0
200
400
600
800